

Math 248 - HWG

$$25. p_n = \frac{1}{3} p_{n-1}, p_1 = \frac{1}{3}$$

$$q_n = 4q_{n-1} - \frac{11}{9} q_{n-2}, q_1 = \frac{1}{3}, q_2 = \frac{1}{9}$$

$$a) \text{ Assume: } p_{n-1} = \left(\frac{1}{3}\right)^{n-1}$$

$$\text{Then: } p_n = \frac{1}{3} \cdot \left(\frac{1}{3}\right)^{n-1} = \left(\frac{1}{3}\right)^n = \left(\frac{1}{3}\right)^n$$

$$\text{Assume: } q_{n-1} = \left(\frac{1}{3}\right)^{n-1}, q_{n-2} = \left(\frac{1}{3}\right)^{n-2}$$

$$\text{Then: } q_n = 4\left(\frac{1}{3}\right)^{n-1} - \frac{11}{9}\left(\frac{1}{3}\right)^{n-2} = \left(\frac{1}{3}\right)^{n-2} \left[4 \cdot \frac{1}{3} - \frac{11}{9}\right]$$

$$= \left(\frac{1}{3}\right)^{n-2} \left[\frac{4}{3} - \frac{11}{9}\right] = \left(\frac{1}{3}\right)^{n-2} \left[\frac{12-11}{9}\right]$$

$$= \left(\frac{1}{3}\right)^{n-2} \cdot \frac{1}{9} = \left(\frac{1}{3}\right)^n = \left(\frac{1}{3}\right)^n$$

$$b) p_{30} = 4.85694 \times 10^{-15} \quad \text{Error: } 1.57772 \times 10^{-30}$$

$\approx \left(\frac{1}{3}\right)^{30}$ (Low Error $\rightarrow$ Good approximation)

$$q_{30} = -1.05458 \times 10^{-1} \quad \text{Error: } 1.05458 \times 10^{-1}$$

(High Error $\rightarrow$ Bad approximation)

$$c) q_n = \frac{33a-9b}{10} \left(\frac{1}{3}\right)^n + \frac{9b-3a}{110} \left(\frac{11}{3}\right)^n$$

$$q_n = A\left(\frac{1}{3}\right)^n + B\left(\frac{11}{3}\right)^n$$

$$r^2 - 4r + \frac{11}{9} = 0$$

$$r = \frac{4 \pm \sqrt{16 - 4 \cdot \frac{11}{9}}}{2} = \frac{4 \pm \sqrt{\frac{144-44}{9}}}{2} = \frac{4 \pm \sqrt{\frac{100}{9}}}{2} = \frac{4 \pm \frac{10}{3}}{2}$$

$$r_1 = \frac{11}{3}, r_2 = \frac{1}{3}$$

$$q_1 = A\left(\frac{1}{3}\right)^1 + B\left(\frac{11}{3}\right)^1 = A + \frac{11B}{3} = \frac{1}{3} \rightarrow A + 11B = 1$$

$$q_2 = A\left(\frac{1}{3}\right)^2 + B\left(\frac{11}{3}\right)^2 = \frac{A}{9} + \frac{121B}{9} = \frac{1}{9} \rightarrow A + 121B = 1$$

Math 248 - HW 6

▷ 25c. $(A+12B) - (A+11B) = 10B = 0 \rightarrow B=0$

▷ $A+11(0) = 1 \rightarrow A=1$

▷ $q_n = A\left(\frac{1}{3}\right)^n + B\left(\frac{1}{3}\right)^n$

▷ $q_n = 1\left(\frac{1}{3}\right)^n + 0\left(\frac{1}{3}\right)^n = \left(\frac{1}{3}\right)^n$

▷ The recurrence relation q is bad because $\left(\frac{1}{3}\right)^n$ grows exponentially fast. This unstable second term dominates quickly leading to huge errors.

▷ 27. $f(x) = 273000x^4 - 277490x^3 - 228731x^2 + 256181x - 31234$

▷ a) Bisection: $x_1 = -0.9700000000$

▷ $f(x) = 0$ $x_2 = 0.1428571428$

▷ $x_3 = 0.7666666666$

▷ $x_4 = 1.0769230769$

▷ b) Newton's Method: $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$

▷ $x_1 = -1$ (Starting value 1)

▷ $x_2 = x_1 - \frac{f(x_1)}{f'(x_1)} = -1 - \frac{f(-1)}{f'(-1)} = -0.9716359149$

▷ $x_3 = x_2 - \frac{f(x_2)}{f'(x_2)} = -0.9716359149 - \frac{f(-0.9716359149)}{f'(-0.9716359149)} = -0.97000$

▷ 3 steps to find with $x_1 = -1$

Math 248 - HW6

27b. $x_1 = 0$ (Starting value 2)

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)} = 0 - \frac{f(0)}{f'(0)} = 0.1219216101$$

$$x_3 = x_2 - \frac{f(x_2)}{f'(x_2)} = 0.1219216101 - \frac{f(0.12...)}{f'(0.12...)} = 0.1421448811$$

$$x_4 = x_3 - \frac{f(x_3)}{f'(x_3)} = 0.1421448811 - \frac{f(0.14...)}{f'(0.14...)} = 0.1428562451$$

$$x_5 = x_4 - \frac{f(x_4)}{f'(x_4)} = 0.1428562451 - \frac{f(0.14...)}{f'(0.14...)} = 0.1428571428$$

5 steps to find with $x_1 = 0$

$x_1 = 0.5$ (Starting value 3)

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)} = 0.5 - \frac{f(0.5)}{f'(0.5)} = 0.9992358634$$

$$x_3 = x_2 - \frac{f(x_2)}{f'(x_2)} = 0.9992358634 - \frac{f(0.99...)}{f'(0.99...)} = 1.1442302919$$

$$x_4 = x_3 - \frac{f(x_3)}{f'(x_3)} = 1.1442302919 - \frac{f(1.14...)}{f'(1.14...)} = 1.0915360971$$

$$x_5 = x_4 - \frac{f(x_4)}{f'(x_4)} = 1.0915360971 - \frac{f(1.09...)}{f'(1.09...)} = 1.0778467411$$

$$x_6 = x_5 - \frac{f(x_5)}{f'(x_5)} = 1.0778467411 - \frac{f(1.07...)}{f'(1.07...)} = 1.0769271298$$

$$x_7 = x_6 - \frac{f(x_6)}{f'(x_6)} = 1.0769271298 - \frac{f(1.07...)}{f'(1.07...)} = 1.076923077$$

7 steps to find with $x_1 = 0.5$

Math 248 - HW6

▷ 27b. $x_1 = 0.75$ (Starting Value 4)

▷ $x_2 = x_1 - \frac{f(x_1)}{f'(x_1)} = 0.75 - \frac{f(0.75)}{f'(0.75)} = 0.7664391571$

▷ $x_3 = x_2 - \frac{f(x_2)}{f'(x_2)} = 0.7664391571 - \frac{f(0.76\dots)}{f'(0.76\dots)}$

▷ $= 0.7666666127$

▷ $x_4 = x_3 - \frac{f(x_3)}{f'(x_3)} = 0.7666666127 - \frac{f(0.76\dots)}{f'(0.76\dots)}$

▷ $= 0.7666666666$

▷ (4 Steps to find with $x_1 = 0.75$)

▷ 27c. $x = -0.9700000000$

▷ 6 Steps with $x_0 = -2, x_1 = -1$

▷ $x = 0.1428571428$

▷ 8 Steps with $x_0 = -1, x_1 = 0$

▷ $x = 0.7666666666$

▷ 11 Steps with $x_0 = 0, x_1 = 0.5$

▷ $x = 1.0769230769$

▷ 10 Steps with $x_0 = 1, x_1 = 2$

▷ 28. $f(x) = x^3 - x^2$ has 2 roots: $(0,0)$ and $(1,0)$.

▷ Finding $(1,0)$ via bisection would be fine because it crosses the x-axis, however $f(x)$ does not cross the x-axis at $(0,0)$ so the method of bisection would not work because it would fail the IVT. ($f(x) < 0$ on both sides of $(0,0)$)

Math 248 - HW6

$$29. f(x) = \frac{1}{x} - c = 0$$

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

$$f'(x) = -\frac{1}{x^2}$$

$$x_{n+1} = x_n - \frac{\frac{1}{x_n} - c}{-\frac{1}{x_n^2}} = x_n + \left(\frac{\frac{1}{x_n} - c}{\frac{1}{x_n^2}} \right) = x_n + (1 - cx_n)x_n$$

$$= x_n(2 - cx_n)$$

$$c = 7, x_0 = 0.1 \quad x = \frac{1}{7} \approx 0.1428571428$$

$$x_1 = x_0(2 - 7x_0) = 0.1(2 - 7(0.1)) = 0.13$$

$$x_2 = x_1(2 - 7x_1) = 0.1417$$

$$x_3 = x_2(2 - 7x_2) = 0.14284777$$

$$x_4 = x_3(2 - 7x_3) = 0.1428571422$$

$$x_5 = x_4(2 - 7x_4) = 0.1428571428 \approx \frac{1}{7}$$

5 Steps With $c=7$ and $x_0=0.1$